SkinSmith: A Constraint-Aware Generative Agent for Game Weapon Skin Design

Registered Project Topic: Latent Vision → Image Generation

YUAN Ye[1] · CHEN Yuhong[1] · Ben Tangjie[1] · Wang Zhengye[1] · WANG Lihui[1]

[1] Master of Science in Data Science, City University of Hong Kong

Supervisor: Prof. Yu Yang | SDSC6002 Research Project | Repository: github.com/DID-FIELD/SDSC6002-SkinSmith

Introduction

- **Background:**
  A visually coherent source image can fail after UV mapping because the atlas fragments, rotates, and repacks the weapon surface. Source-only assessment can therefore hide internal seams, lost motifs, and poor top-view readability.
- **Objective:**
  Develop and evaluate a complete image-processing/generation Agent that converts one short theme keyword into a human-selected, constraint-valid game-asset texture through three explicit checkpoints.

Methodology

The proposed workflow combines semantic planning, human checkpoints, asset-aware generation, render-in-the-loop evaluation, and one bounded correction with rollback.

Figure 1. Continuous weapon-space composition → authoritative UV atlas → reconstructed 3D weapon.

- **Route A — PatternDesigner:**
  A deterministic or generated tileable pattern baseline with periodic-boundary repair.
- **Route B — WeaponThemeDesigner:**
  One dense master artwork is fitted in continuous weapon space and baked through OBJ-derived UV maps.
- **Route C — Bounded refinement:**
  A render-conditioned local action is accepted only when hard constraints pass and the locked score improves.

Human–Agent Checkpoints

- **Checkpoint 1 — Theme world:**
  Select a weapon and enter one keyword. Confirm or edit subjects, symbols, materials, palette, atmosphere, and prohibited content.
- **Checkpoint 2 — Art direction:**
  Choose one of 3–4 materially different textual directions before image generation.
- **Checkpoint 3 — Mapped artwork:**
  Choose one source + left/right/top candidate card. Model review is recommendation-only; the human decision is authoritative.

Experimental Setup

- **Case assets:**
  AK-47 complete end-to-end case; M4A4 adapter transfer case.
- **Primary metrics:**
  Asset-seam error, multi-view score, texture score, feasibility thresholds, locality, and rollback decision.
- **Evaluation design:**
  Four paired Route A/B candidates; edge-width and selection-policy sub-ablation; preserved live Gemini run.

Results & Analysis

- **Superior asset feasibility:**
  Route B reduced mean asset-seam error from 0.26997 to 0.00063 — a 99.77% reduction — and every paired candidate passed the 0.01 threshold.
- **Mapped-view improvement:**
  Mean multi-view score increased from 0.76898 to 0.82911 (+0.060); all 4/4 paired candidates improved both primary outcomes.
- **Constraint-first selection:**
  A weighted score chose 4 px despite a seam violation. Constraint-first + Pareto selected the first jointly feasible point at 8 px.
- **Safe rollback:**
  The live Route C correction reduced score by 0.02196, so the Agent retained Route B rather than presenting a harmful refinement.

Figure 2a. Route A pattern-first baseline.

Figure 2b. Route B weapon-space result.

Figure 3a. Human-selected Treasured Relic source.

Figure 3b. Formal left/right/top mapped result.

Formal paired result: 99.77% mean seam reduction | +0.060 mean multi-view score | 4/4 paired wins
Live selected result: seam 0.00078 | multi-view 0.8166 | total score 0.8291 | Route C rejected

Conclusion

SkinSmith shows that a generated rectangle is not a valid asset until it is mapped, inspected from multiple views, and tested against deployment constraints. The project contributes asset-aware weapon-space generation, mapped-view human selection, and bounded refinement with safe rollback. AK-47 is the complete case; M4A4 demonstrates adapter transfer rather than universal parameter portability.